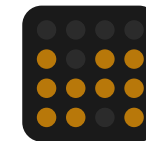


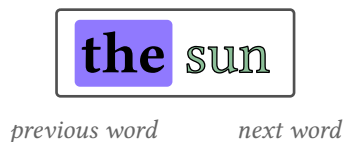
How to use these token cutouts



These pages contain the text *The Cat in the Hat* by Dr Seuss. Each **cutout** shows a **next word** paired with the **previous word** that came immediately before it in the original text.

Model vitals	
1962 tokens	2.83 bits/token entropy
7.1 perplexity	4.32 branching factor

Anatomy of a cutout



Every distinct word has its own colour. **Previous** words appear inside a coloured box (the word’s own colour as the background, with white text); the free-standing **next word** appears in plain coloured text. The same word always wears the same colour, whether you see it inside a box or free-standing. Two unrelated words can occasionally share a colour, so always verify the word itself matches—not just the colour.

Setup

Cut out the tokens along the dotted lines and **spread them out** face-up on the table, with no overlap if possible. This is the “training” step: every (previous, next) combination from the original text is now a physical cutout sitting on your table.

How to play: the matching game

To generate text, repeatedly find a cutout whose **previous word** matches the last word you’ve written, then **write its next word** onto your page. The word you just wrote becomes part of what you’ll match against next—that’s the chain.

Worked example

Step 1. Pick any cutout to begin—say this one—and copy its previous word *and* its next word onto your page.

Step 2. Look at the last word you’ve written (*sun*) and find a cutout whose previous word matches it. Write its next word.

Step 3. Repeat: find a cutout whose previous word matches the last word you’ve written, and add its next word.

Keep going—write as much or as little as you like.

the cutout

the sun

sun did

did not

your page

the sun

the sun **did**

the sun did **not**

Tips

Use colour as a fast filter. Scan by the rightmost previous-word box’s colour first, then verify the actual word matches.

Pick from many candidates by eye. The more cutouts share the same previous word, the more often your eye lands on one—that’s **weighted sampling** for free, and it’s why some words follow others more often in your text.

Put the cutout back after using it—removing it would change the model’s distribution next time round.

the **the** sun **sun** did **did** not **not** shine **shine** .

. it **it** was **was** too **too** wet **wet** to **to** play

play . . so **so** we **we** sat **sat** in **in** the

the house **house** all **all** that **that** cold **cold** ,

, cold **cold** , , wet **wet** day **day** . . I **I** sat

sat there **there** with **with** Sally **Sally** . . we

we sat **sat** there **there** , , we **we** two **two** .

. and **and** I **I** said **said** , , how **how** I **I** wish

wish we **we** had **had** something **something** to

to do **do** ! ! too **too** wet **wet** to **to** go **go** out

out and **and** too **too** cold **cold** to **to** play

play ball **ball** . **.** so **so** we **we** sat **sat** in **in** the

the house **house** . **.** we **we** did **did** nothing

nothing at **at** all **all** . **.** so **so** all **all** we

we could **could** do **do** was **was** to **to** sit **sit** !

! sit **sit** ! **!** sit **sit** ! **!** sit **sit** ! **!** and **and** we

we did **did** not **not** like **like** it **it** . **.** not

not one **one** little **little** bit **bit** . **.** and **and** then

then something **something** went **went** bump

bump ! **!** how **how** that **that** bump **bump** made

made us **us** jump **jump** ! **!** we **we** looked

looked ! **!** then **then** we **we** saw **saw** him

him step **step** in **in** on **on** the **the** mat **mat** !

! we **we** looked **looked** ! **!** and **and** we **we** saw

saw him **him** ! **!** the **the** cat **cat** in **in** the

the hat **hat** ! **!** and **and** he **he** said **said** to

to us **us** , **,** Why **Why** do **do** you **you** sit

sit there **there** like **like** that **that** ? **?** I **I** know

know it **it** is **is** wet **wet** and **and** the **the** sun

sun is **is** not **not** sunny **sunny** . **.** But **But** we

we can **can** have **have** Lots **Lots** of **of** good

good fun **fun** that **that** is **is** funny **funny** ! **!** I

I know **know** some **some** good **good** games

games **we** **we** could **could** play **play** , **,** said

said the **the** cat **cat** . **.** I **I** know **know** some

some new **new** tricks **tricks** , **,** said **said** the

the cat **cat** in **in** the **the** hat **hat** . **.** a **a** lot

lot of **of** good **good** tricks **tricks** . **.** I **I** will

will show **show** them **them** to **to** you **you** .

. your **your** mother **mother** will **will** not

not mind **mind** at **at** all **all** if **if** I **I** do **do** .

. **then** **then** Sally **Sally** and **and** I **I** did **did** not

not know **know** what **what** to **to** say **say** .

. our **our** mother **mother** was **was** out **out** of

of the **the** house **house** For **For** the **the** day

day . . But **But** our **our** fish **fish** said **said** ,

, no **no** ! **!** no **no** ! **!** Make **Make** that **that** cat

cat go **go** away **away** ! **!** tell **tell** that **that** cat

cat in **in** the **the** hat **hat** you **you** do **do** not

not want **want** to **to** play **play** . . he **he** should

should not not be be here here . he

he should should not not be be about about .

. he he should should not not be be here

here When When your your mother mother is

is out out ! ! now now ! ! now now ! ! have

have no no fear fear . . have have no no fear

fear ! ! said said the the cat cat . . my

my tricks tricks are are not not bad bad ,

, said said the the cat cat in in the the hat

hat . . Why Why , , we we can can have

have Lots Lots of of good good fun fun , if
if you you wish wish , with with a a game
game that that I I call call up up up up
up with with a a fish fish ! ! put put me
me down down ! ! said said the the fish fish .
. this this is is no no fun fun at at all all !
! put put me me down down ! ! said said the
the fish fish . . I I do do not not wish
wish to to fall fall ! ! have have no no fear
fear ! ! said said the the cat cat . . I I will

will not **not** let **let** you **you** fall **fall** . **.** I **I** will

will hold **hold** you **you** up **up** high **high** as

as I **I** stand **stand** on **on** a **a** ball **ball** . **.** with

with a **a** book **book** one **one** one **one** hand

hand ! **!** and **and** a **a** cup **cup** on **on** my

my hat **hat** ! **!** But **But** that **that** is **is** not

not all **all** I **I** can **can** do **do** ! **!** said **said** the

the cat **cat** . **.** **.** **.** look **look** at **at** me **me** !

! look **look** at **at** me **me** now **now** ! **!** said

said the **the** cat **cat** . **.** with **with** a **a** cup

cup and **and** a **a** cake **cake** on **on** the **the** top

top of **of** my **my** hat **hat** ! **!** I **I** can **can** hold

hold up **up** two **two** books **books** ! **!** I **I** can

can hold **hold** up **up** the **the** fish **fish** ! **!** and

and a **a** little **little** toy **toy** ship **ship** ! **!** and

and some **some** milk **milk** on **on** a **a** dish

dish ! **!** and **and** look **look** ! **!** I **I** can **can** hop

hop up **up** and **and** down **down** on **on** the

the ball **ball** ! **!** But **But** that **that** is **is** not

not all **all** ! **!** Oh **Oh** , **,** no **no** . **.** that **that** is

is not **not** all **all** . . . look **look** at **at** me

me ! ! look **look** at **at** me **me** ! ! look **look** at

at me **me** now **now** ! ! it **it** is **is** fun **fun** to

to have **have** fun **fun** But **But** you **you** have

have to **to** know **know** how **how** . I **I** can

can hold **hold** up **up** the **the** cup **cup** and

and the **the** milk **milk** and **and** the **the** cake

cake ! ! I **I** can **can** hold **hold** up **up** these

these books **books** ! ! and **and** the **the** fish

fish on **on** a **a** rake **rake** ! ! I **I** can **can** hold

hold the **the** toy **toy** ship **ship** and **and** a

a little **little** toy **toy** man **man** ! **!** and

and look **look** ! **!** with **with** my **my** tail **tail** I

I can **can** hold **hold** a **a** red **red** fan **fan** ! **!** I

I can **can** fan **fan** with **with** the **the** fan **fan** as

as I **I** hop **hop** on **on** the **the** ball **ball** ! **!** But

But that **that** is **is** not **not** all **all** . **.** Oh **Oh** ,

, no **no** . **.** that **that** is **is** not **not** all **all** . **.**

. **.** that **that** is **is** what **what** the **the** cat

cat said **said** . **.** **.** **.** **.** then **then** he **he** fell

fell on **on** his **his** head **head** ! **!** he **he** came

came down **down** with **with** a **a** bump

bump From **From** up **up** there **there** on **on** the

the ball **ball** . **.** and **and** Sally **Sally** and **and** I

I , **,** we **we** saw **saw** all **all** the **the** things

things fall **fall** ! **!** and **and** our **our** fish

fish came **came** down **down** , **,** too **too** . **.** he

he fell **fell** into **into** a **a** pot **pot** ! **!** he **he** said

said , **,** do **do** I **I** like **like** this **this** ? **?** Oh

Oh , **,** no **no** ! **!** I **I** do **do** not **not** . **.** this

this is **is** not **not** a **a** good **good** game **game** ,

, said **said** our **our** fish **fish** as **as** he **he** lit

lit . **.** no **no** , **,** I **I** do **do** not **not** like **like** it

it , **,** not **not** one **one** little **little** bit **bit** !

! now **now** look **look** what **what** you **you** did

did ! **!** said **said** the **the** fish **fish** to **to** the

the cat **cat** . **.** now **now** look **look** at **at** this

this house **house** ! **!** look **look** at **at** this **this** !

! look **look** at **at** that **that** ! **!** you **you** sank

sank our **our** toy **toy** ship **ship** , **,** sank

sank it it deep deep in in the the cake cake .

. you you shook shook up up our our house

house and and you you bent bent our

our new new rake rake . . you you should

should not not be be here here When

When our our mother mother is is not not .

. you you get get out out of of this this house

house ! ! said said the the fish fish in in the

the pot pot . . But But I I like like to to be

be here here . . Oh Oh , , I I like like it it a

a lot lot ! ! said said the the cat cat in in the
the hat hat to to the the fish fish in in the
the pot pot . . I I will will not not go
go away away . . I I do do not not wish
wish to to go go ! ! and and so so , , said
said the the cat cat in in the the hat hat ,
, so so so so so I I will will show
show you you another another good
good game game that that I I know know !
! and and then then he he ran ran out out .

. and and , then then , fast fast as as a
a fox fox , the the cat cat in in the the hat
hat came came back back in in with with a
a box box . a a big big red red wood
wood box box . it it was was shut shut with
with a a hook hook . now now look look at
at this this trick trick , said said the the cat
cat . take take a a look look ! ! then
then he he got got up up on on top top with
with a a tip tip of of his his hat hat . I

I call call this this game game fun fun in in a
a box box , , said said the the cat cat . . in
in this this box box are are two two things
things I I will will show show to to you
you now now . . you you will will like
like these these two two things things , , said
said the the cat cat with with a a bow bow .
. I I will will pick pick up up the the hook
hook . . you you will will see see something
something new new . . two two things

things . and and I I call call them

them Thing Thing one one and and Thing

Thing two two . these these things

things will will not not bite bite you you .

. they they want want to to have have fun

fun . then then , , out out of of the the box

box came came Thing Thing two two and

and Thing Thing one one ! ! and and they

they ran ran to to us us fast fast . they

they said said , , how how do do you you do

do ? ? would would you you like like to

to shake shake hands hands with with Thing

Thing one one and and Thing Thing two

two ? ? and and Sally Sally and and I I did

did not not know know what what to to do

do . . so so we we had had to to shake

shake hands hands with with Thing Thing one

one and and Thing Thing two two . . we

we shook shook their their two two hands

hands . . But But our our fish fish said said ,

, no no ! ! no no ! ! those those things

things should should not not be be in in this

this house house ! ! Make Make them

them go go ! ! they they should should not

not be be here here When When your

your mother mother is is not not ! ! put

put them them out out ! ! put put them

them out out ! ! said said the the fish fish in

in the the pot pot . . have have no no fear

fear , , little little fish fish , , said said the

the cat **cat** **in** **in** the **the** hat **hat** . **.** these

these things **things** are **are** good **good** things

things . **.** and **and** he **he** gave **gave** them

them a **a** pat **pat** . **.** they **they** are **are** tame

tame . **.** Oh **Oh** , **,** so **so** tame **tame** ! **!** they

they have **have** come **come** here **here** to

to play **play** . **.** they **they** will **will** give

give you **you** some **some** fun **fun** on **on** this

this wet **wet** , **,** wet **wet** , **,** wet **wet** day **day** .

. now **now** , **,** here **here** is **is** a **a** game

game that that they they like like , , said

said the the cat cat . . they they like like to

to fly fly kites kites , , said said the the cat

cat in in the the hat hat . . no no ! ! not

not in in the the house house ! ! said said the

the fish fish in in the the pot pot . . they

they should should not not fly fly kites

kites in in a a house house ! ! they

they should should not not . . Oh Oh , , the

the things things they they will will bump

bump ! ! Oh Oh , , the the things things they

they will will hit hit ! ! Oh Oh , , I I do

do not not like like it it ! ! not not one

one little little bit bit ! ! then then Sally

Sally and and I I saw saw them them run

run down down the the hall hall . . we

we saw saw those those two two things

things bump bump their their kites kites on

on the the wall wall ! ! bump bump !

! Thump Thump ! ! Thump Thump ! ! bump

bump ! ! down down the the wall wall in

in the the hall hall . . Thing Thing two

two and and Thing Thing one one ! ! they

they ran ran up up ! ! they they ran

ran down down ! ! on on the the string

string of of one one kite kite we we saw

saw mother's mother's new new gown gown !

! her her gown gown with with the the dots

dots that that are are pink pink , , white

white and and red red . . then then we

we saw saw one one kite kite bump bump on
on the the head head of of her her bed bed !
! then then those those things things ran
ran about about with with big big bumps
bumps , , jumps jumps and and kicks
kicks and and with with hops hops and
and big big thumps thumps and and all
all kinds kinds of of bad bad tricks tricks .
. and and I I said said , , I I do do not
not like like the the way way that that they

they play **play** ! **!** if **if** mother **mother** could

could see **see** this **this** , **,** Oh **Oh** , **,** what

what would **would** she **she** say **say** ! **!** then

then our **our** fish **fish** said **said** , **,** look **look** !

! look **look** ! **!** and **and** our **our** fish **fish** shook

shook with **with** fear **fear** . **.** your **your** mother

mother is **is** on **on** her **her** way **way** home

home ! **!** do **do** you **you** hear **hear** ? **?** Oh

Oh , **,** what **what** will **will** she **she** do **do** to

to us **us** ? **?** what **what** will **will** she **she** say

say ? ? Oh Oh , , she she will will not

not like like it it to to find find us us this

this way way ! ! so so do do something

something ! ! fast fast ! ! said said the

the fish fish . . do do you you hear hear ! ! I

I saw saw her her . . your your mother

mother ! ! your your mother mother is is near

near ! ! so so , , as as fast fast as as you

you can can , , Think Think of of something

something to to do do ! ! you you will

will have have to to get get rid rid of
of Thing Thing one one and and Thing
Thing two two ! ! so so , , as as fast fast as
as I I could could , , I I went went after
after my my net net . . and and I I said
said , , with with my my net net I I can
can get get them them I I bet bet . . I I bet
bet , , with with my my net net , , I I can
can get get those those things things yet yet !
! then then I I let let down down my my net

net . it it came came down down with

with a a PLOP PLOP ! ! and and I I had

had them them ! ! at at last last ! ! those

those two two things things had had to

to stop stop . . then then I I said said to

to the the cat cat , , now now you you do

do as as I I say say . . you you pack pack up

up those those things things and and you

you take take them them away away ! ! Oh

Oh dear dear ! ! said said the the cat cat .

. you you did did not not like like our

our game game Oh Oh dear dear .

. what what a a shame shame ! ! what what a

a shame shame ! ! what what a a shame

shame ! ! then then he he shut shut up

up the the things things in in the the box

box with with the the hook hook . . and

and the the cat cat went went away

away with with a a sad sad kind kind of

of look look . . that that is is good good ,

, said said the the fish fish . he he has

has gone gone away away . Yes Yes . But

But your your mother mother will will come

come . she she will will find find this

this big big mess mess ! ! and and this

this mess mess is is so so big big and and so

so deep deep and and so so tall tall , , we

we can can not not pick pick it it up up .

. there there is is no no way way at at all

all ! ! and and then then ! ! Who Who was

was back back in in the the house house ?

? Why Why , , the the cat cat ! ! have

have no no fear fear of of this this mess

mess , , said said the the cat cat in in the

the hat hat . . I I always always pick pick up

up all all my my playthings playthings and

and so so I I will will show

show you you another another good good trick

trick that that I I know know ! ! then

then we we saw saw him him pick pick up

up all all the the things things that that were
were down down . he he picked picked up
up the the cake cake , , and and the the rake
rake , , and and the the gown gown , , and
and the the milk milk , , and and the
the strings strings , , and and the the books
books , , and and the the dish dish , , and
and the the fan fan , , and and the the cup
cup , , and and the the ship ship , , and
and the the fish fish . and and he he put

put them them away away . then then he
he said said , , that that is is that that . and
and then then he he was was gone gone with
with a a tip tip of of his his hat hat . then
then our our mother mother came came in
in and and she she said said to to us us two
two , , did did you you have have any
any fun fun ? ? tell tell me me . what
what did did you you do do ? ? and and Sally
Sally and and I I did did not not know

know what **what** to **to** say **say** . **.** should

should we **we** tell **tell** her **her** the **the** things

things that **that** went **went** on **on** there

there that **that** day **day** ? **?** should **should** we

we tell **tell** her **her** about **about** it **it** ? **?** now

now , **,** what **what** should **should** we **we** do

do ? **?** Well **Well** . **.** **.** **.** what **what** would

would you **you** do **do** if **if** your **your** mother

mother asked **asked** you **you** ?